Thanmay M Shetty

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Summary

AI Engineer with experience building agentic AI systems, multilingual RAG pipelines, AI-powered automation tools, and full-stack applications across healthcare, agri-tech, scientific data, and NLP domains. Developed production-ready solutions using Python, FastAPI, LangChain, vector databases, and modern LLM APIs while improving query accuracy by 20% and model accuracy from 76% to 94%. Top-3 finisher across 7 national and inter-collegiate hackathons; managed a 15-member technical chapter with a ₹50,000 annual budget.

Experience

PGAGI — AI/ML Intern | Jul 2026 – Present

- **Build and support production AI features for a product serving 10K+ users using FastAPI, GCP, Firestore, Docker, Git/GitHub, logging, caching, API design, testing, and cron jobs.**
- **Develop end-to-end RAG/LLM pipelines using Claude, OpenAI, and OpenRouter; evaluate RAGAS and integrate text-to-speech, image-generation, and video-generation models while optimizing inference cost.**

Education

B.E. in Artificial Intelligence & Machine Learning *2027*
BMS Institute of Technology and Management (BMSITM), Bengaluru CGPA: 9.01
PUC in PCMC 2023
Christ Academy Junior College ,Bengaluru 95%

Technical Skills

Languages: Python, JavaScript, Java, C, SQL
Computer Vision & Deep Learning: OpenCV, Image Processing, Face Recognition, Object Detection, Image Classification, PyTorch, TensorFlow
AI/ML: PyTorch, TensorFlow, Scikit-learn, Deep Learning, Machine Learning, GenAI, Agentic AI, LLMs, Model Training, Model Evaluation
Frameworks: FastAPI, Next.js, React, Node.js, REST APIs, LangChain, PyTorch, TensorFlow
Machine Learning: Model Training, Evaluation, Data Augmentation, Neural Networks, Transformers, Pandas, NumPy, Matplotlib, Seaborn
DevOps & Tools: Firestore, GCP, Docker, Docker Compose, GitHub Actions, CI/CD, Linux, Git/Github, Logging, Caching, Cron Jobs
Algorithms & DS: Data structures, algorithm design, complexity analysis

Project Experience

BioBloom Ecosystem — 1st Place, RNSIT National Hackathon

- **Architected** sole-developed **agentic RAG** pipeline (AgriGain); indexed **300+ knowledge chunks**, implemented **3-second API timeout** guards, integrated cloud database, backend, and frontend; deployed to Vercel.
- **Generated** 4 comparative waste utilization strategies per input across 7 metrics with 90% structured output success rate; delivered results in English, Hindi, and Kannada.
- **Curated** 3GB of crop rotation research papers (AgriKami) as knowledge base for a RAG-based crop planning system.

ARGO NETCDF Explorer — Scientific Data RAG Pipeline

- **Constructed** RAG pipeline from scratch over **5GB of oceanographic data**; parsed, chunked, and embedded **~20,000 profile chunks** with structured metadata into a vector database with filtered retrieval.
- **Improved** natural-language query accuracy from ~70% to ~90% across 5 prompt engineering iterations; built interactive dashboard with 6–7 data widgets including location maps and multi-year trend graphs.

Sentiment Analysis API — DistilBERT

- **Architected** a production-grade NLP inference platform by fine-tuning **DistilBERT** on **3.6M** Amazon reviews, improving classification accuracy from **76%** to **94%**; deployed a **FastAPI** inference service with **~500 ms** latency using **Docker**, **Azure Container Registry (ACR)**, and **Azure Container Apps**.
- **Engineered** a cloud-native microservices architecture using **Azure Application Gateway**, **Nginx** reverse proxy, **Azure Virtual Network (VNet)**, **Azure Virtual Machine-hosted PostgreSQL**, and **Azure Files**, enabling secure request routing, scalable deployments, and persistent shared storage.
- **Automated** end-to-end **GitHub Actions CI/CD** pipelines with container image versioning, **Azure Container Apps** revision deployments, and **Terraform**-based Infrastructure as Code, validating deployment workflows locally using **Kubernetes (Minikube)**.

Quantum-AI Cognitive Health System — Solo Exploration

- **Architected a real-time AI decision platform processing EEG and HRV streams across 24 REST APIs, classifying 3 cognitive states and dynamically recommending personalized learning tasks.**

Certifications

- **Introduction to LLM** — NPTEL
- **Leadership and Team Effectiveness** — Scored in top 5% nationally

Achievements

- **1st Place** — RNSIT National Hackathon, 24-hour AI/ML hackathon (2025)
- **3rd Place** — KSEEM National Level Hackathon (2025)
- **2nd Place** — Build and Conquer Inter-Collegiate Hackathon (2024)
- **Ranked** in top 3 across 7 hackathons spanning Agri-tech, health-tech, and full-stack AI domains.

Leadership & Other

Treasurer — IEEE Computational Intelligence Society (CIS) *2024 – 2025*
Managed ₹45,000–50,000 annual budget for a 15-member IEEE CIS chapter across 6 events: 3 technical workshops (50–70 attendees each), an inter-collegiate hackathon/ideathon with 50+ teams, and a Winter of Projects mentoring 4 juniors on AI/ML; served as technical spokesperson for LLM and RAG system demos to non-technical stakeholders.